

Question	Answer	Marks
2(a)	in (rotational) equilibrium	B1
	<u>sum / total</u> of CW moments about a point = <u>sum / total</u> of ACW moments about the (same) point.	B1
2(b)(i)	The magnitudes of the moments about A are: (1700 × 3.0) (660 x) (1300 × 5) Correct magnitude of any one moment about A.	C1
	Correct magnitudes of a second moment about A.	C1
	(1700 × 3.0) + (660 x) = (1300 × 5) x = 2.1 m	A1

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2(b)(ii)	(Upthrust =) $11 \times 9.81 + 1300 = 1400$ (N)	A1
2(b)(iii)	$y = \text{Upthrust} / \rho g A$ $= 1400 / (990 \times 9.81 \times \pi \times (0.78 / 2)^2)$	C1
	$= 0.30$ m	A1
	OR $[P = F / A = 1400 / \pi \times (0.78 / 2)^2 = 2930]$ $y = P / \rho g$ $= 2930 / (990 \times 9.81)$	(C1)
	$= 0.30$ m	(A1)
2(b)(iv)	line with a non-zero value of depth at distance = 0	B1
	A line from distance = 0 to distance = 6.0 m with a gradient that is always positive	B1

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